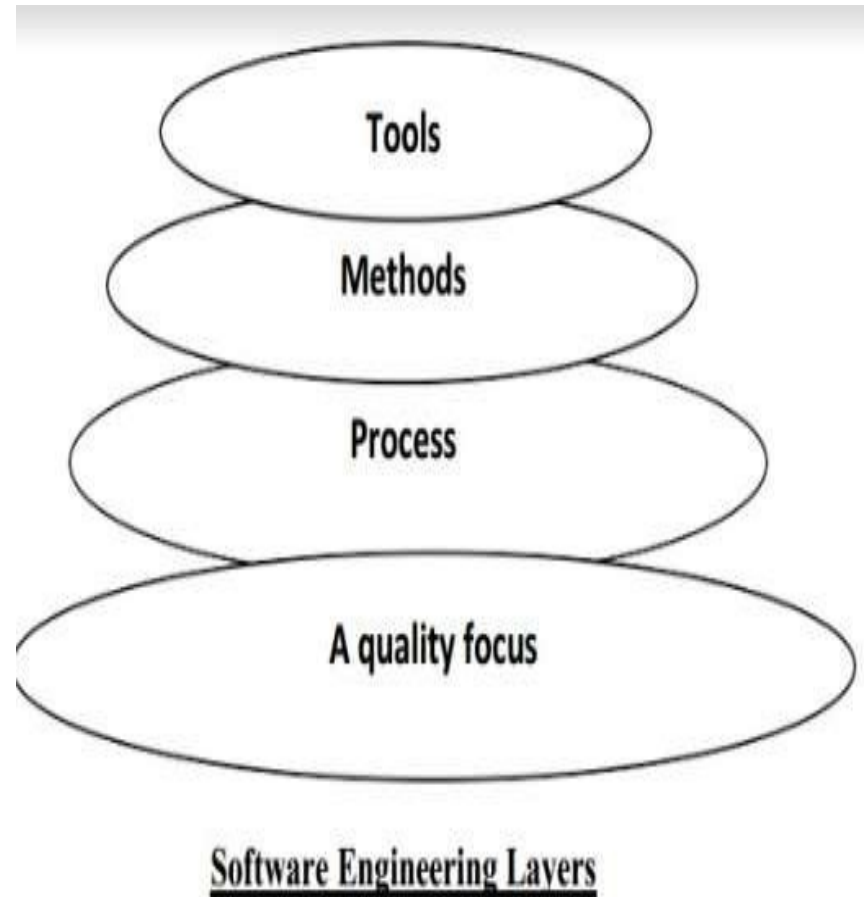


Software Estimation and Scheduling -III

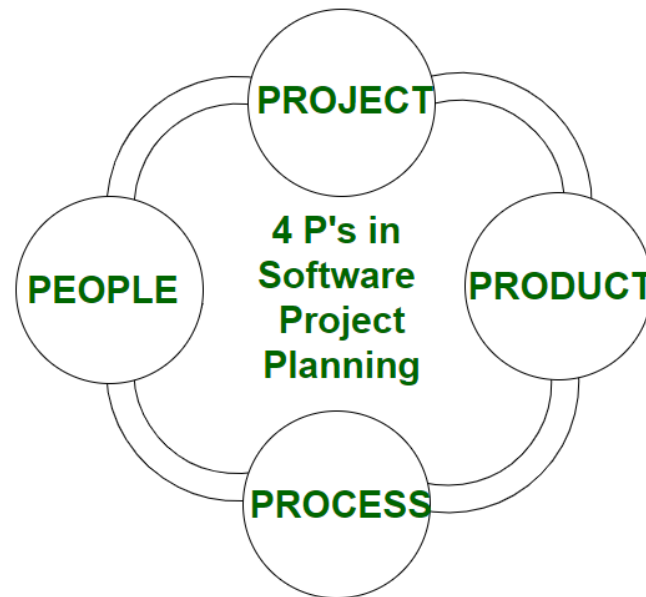
Software Engineering -A layered Technology:

- Application of a systematic, disciplined, quantifiable approach to the development, operation and maintenance of software that is, the application of engineering software.



The Management Spectrum 4 Ps in Software Project Planning

- For properly building a product there are 4 critical components in software project planning which are known as the **4P's** namely:
- Product
- Process
- People
- Project

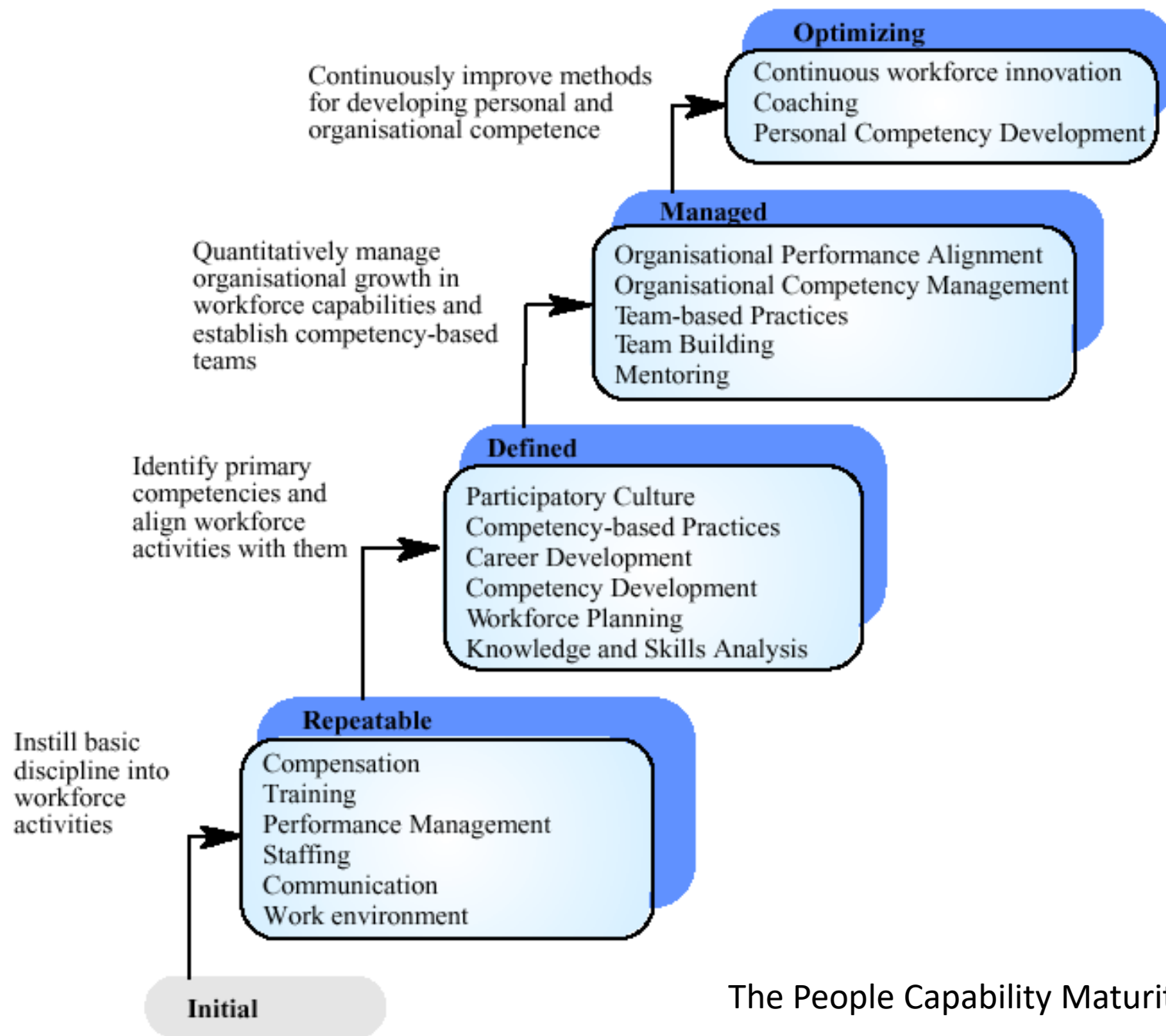


People

- The most important component of a product and its successful implementation is human resources.
- In building a proper product, a well-managed team with clear-cut roles defined for each person/team will lead to the success of the product.
- There has to be a good team in order to save time, cost, and effort.
- Some assigned roles in software project planning are **project manager, team leaders, stakeholders, analysts**, and other **IT professionals**.
- Managing people successfully is a tricky process which a good project manager can do.

The People Capability Maturity Model

- Intended as a framework for managing the development of people involved in software development
- Five stage model
 - Initial. Ad-hoc people management
 - Repeatable. Policies developed for capability improvement
 - Defined. Standardised people management across the organisation
 - Managed. Quantitative goals for people management in place
 - Optimizing. Continuous focus on improving individual competence and workforce motivation



The People Capability Maturity Model

P-CMM Objectives

- To improve organisational capability by improving workforce capability
- To ensure that software development capability is not reliant on a small number of individuals
- To align the motivation of individuals with that of the organisation
- To help retain people with critical knowledge and skills

Key points

- Managers must have some understanding of human factors to avoid making unrealistic demands on people
- Problem solving involves integrating information from long-term memory with new information from short-term memory
- Staff selection factors include education, domain experience, adaptability, and personality

Key points

- Software development groups should be small and cohesive
- Group communications are affected by status, group size, group organisation and the gender composition of the group
- The working environment has a significant effect on productivity
- The People Capability Maturity Model is a framework for improving the capabilities of staff in an organisation

Product

- This is the deliverable or the result of the project.
- The project manager should clearly define the product scope to ensure a successful result, control the team members, as well technical hurdles that he or she may encounter during the building of a product.

Process

- In every planning, a clearly defined process is the key to the success of any product.
- It regulates how the team will go about its development in the respective time period.
- The Process has several steps involved like,
 - documentation phase
 - implementation phase
 - deployment phase
 - interaction phase.

Project

- The last and final P in software project planning is Project. It can also be considered as a blueprint of process.
- In this phase, the project manager plays a critical role.
- They are responsible to guide the team members to achieve the project's target and objectives, helping & assisting them with issues, checking on cost and budget, and making sure that the project stays on track with the given deadlines.